Introduction

This report presents a comprehensive project management analysis of the Learning Management System (LMS), a startup initiative from EduLearn with the aim of transforming online education in Nigeria. EduLearn seeks to address challenges commonly faced by Nigerian universities such as limited access to learning materials, inefficient assignment handling, and weak student-lecturer engagement outside the classroom.

Project Objective:

• **User Authentication System**: Multi -tiered access for students, lecturers,

and administrators with role -based permissions

• **Course Managemen**t: Comprehensive tools for course creation, enrollment,

and content delivery

• **Assignment Workflow**: Submission, grading, and feedback mechanisms

• **Communication Platform**: Discussion forums for collaborative learning

• **Notification System**: Real -time alerts for assignments, announcements, and grades.

Project Scope

The EduLearn LMS project aims to deliver a web -based Learning Management System for Nigerian universities. The system will support:

• User authentication (Students, Lecturers, Admins)

• Course creation and management

• Assignment submission and grading

• Discussion forums

• Real -time notifications

<u>Out of scope (Initial phase):</u>

• Video conferencing

PROJECT COSTRAINTS

Project Duration: 3 months

This three-month duration covers the core planning, design, and development activities documented in this report, while the full project lifecycle spans six months including post-deployment support and maintenance.

Total Budget: ₦ 4,000,000 (adjusted from ₦ 5,000,000)

Team size: 6 members

Work Breakdown Structure (WBS)

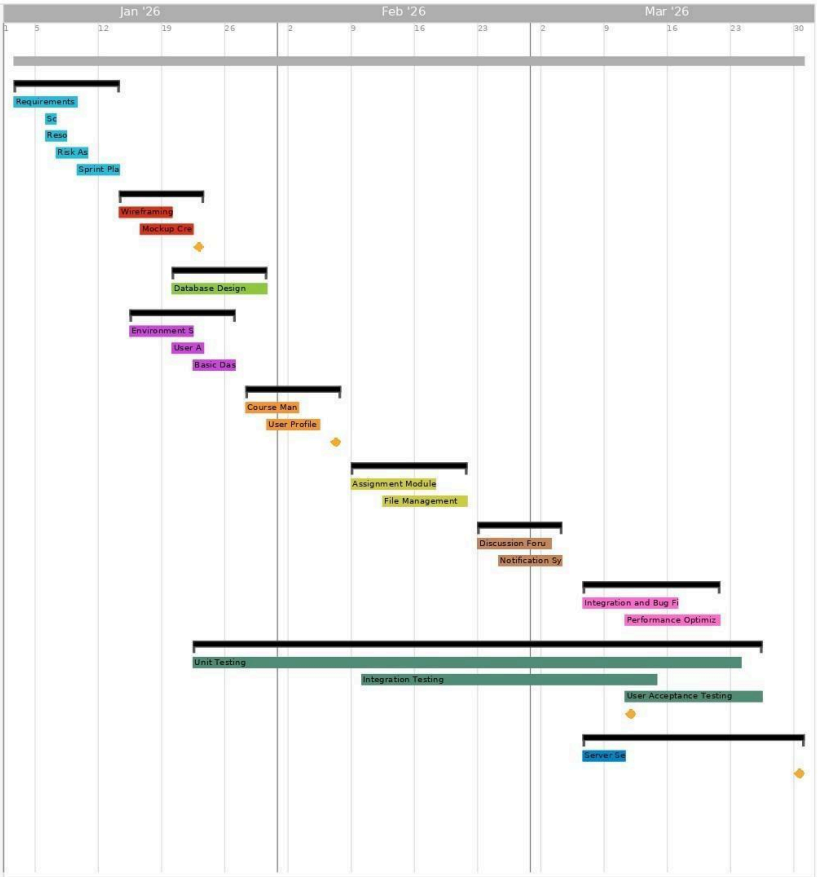
The Work Breakdown Structure (WBS) below outlines the major activities of the project. Detailed design artefacts, including UI wireframes, system architecture, and database design, are provided in the Appendix.

1. Project Management
    - 1.1. Planning
        - 1.1.1. Requirements Gathering
        - 1.1.2. Scope Definition
        - 1.1.3. Resource Planning
        - 1.1.4. Risk Assessment
        - 1.1.5. Sprint Planning
2. Design
    - 2.1. UI/UX Design
        - 2.1.1. Wireframing
        - 2.1.2. Mockup Creation
        - 2.1.3. Prototype Development
        - 2.1.4. Design System Creation
    - 2.2. Architecture
        - 2.2.1. Technology Stack selection
        - 2.2.2. Database Design
        - 2.2.3. API Design
        - 2.2.4. Security Architecture
3. Development
    - 3.1. Sprint 1
        - 3.1.1. Environment Setup
        - 3.1.2. User Authentication
        - 3.1.3. Basic Dashboard
    - 3.2. Sprint 2
        - 3.2.1. Course Management
        - 3.2.2. User Profile Management
    - 3.3. Sprint 3
        - 3.3.1. Assignment Module
        - 3.3.2. File Management

3.4. Sprint 4
    3.4.1. Discussion Forums
    3.4.2. Notification System
3.5. Sprint 5
    3.5.1. Integration and Bug fixes
    3.5.2. Performance Optimization

4. Testing
    4.1. Unit Testing
    4.2. Integration Testing
    4.3. User Acceptance Testing
    4.4. Performance Testing
    4.5. Security Testing
5. Deployment
    5.1. Server Setup
    5.2. Launch
    5.3. Support

EduLearn LMS Project	start	end	100%	
Planning	**02/01/26**	**13/01/26**	**100%**	**10**
Requirements Gathering	02/01	08/01	100%	6 days
Scope Definition	06/01	06/01	100%	1 days
Resource Planning	06/01	07/01	100%	2 days
Risk Assessment	07/01	09/01	100%	3 days
Sprint Planning	09/01	13/01	100%	4 days
UI/UX Design	**14/01/26**	**22/01/26**	**100%**	**8**
Wireframing	14/01	19/01	100%	5 days
Mockup Creation	16/01	21/01	100%	5 days
UI/UX Design Approval	22/01	22/01	100%	1 days
Architecture	**20/01/26**	**29/01/26**	**100%**	**9**
Database Design	20/01	29/01	100%	9 days
Sprint 1	**15/01/26**	**26/01/26**	**100%**	**10**
Environment Setup	15/01	21/01	100%	6 days
User Authentication	20/01	22/01	100%	3 days
Basic Dashboard	22/01	26/01	100%	4 days
Sprint 2	**28/01/26**	**06/02/26**	**100%**	**9**
Course Management	28/01	02/02	100%	5 days
User Profile Management	30/01	04/02	100%	5 days
Sprint 2 Demo	06/02	06/02	100%	1 days
Sprint 3	**09/02/26**	**20/02/26**	**100%**	**11**
Assignment Module	09/02	17/02	100%	8 days
File Management	12/02	20/02	100%	8 days
Sprint 4	**23/02/26**	**03/03/26**	**100%**	**8**
Discussion Forums	23/02	02/03	100%	7 days
Notification System	25/02	03/03	100%	6 days
Sprint 5	**06/03/26**	**20/03/26**	**100%**	**13**
Integration and Bug Fixes	06/03	16/03	100%	9 days
Performance Optimization	11/03	20/03	100%	9 days
Testing	**22/01/26**	**25/03/26**	**100%**	**54**
Unit Testing	22/01	23/03	100%	52 days
Integration Testing	10/02	13/03	100%	28 days
User Acceptance Testing	11/03	25/03	100%	13 days
UAT Start	11/03	11/03	100%	1 days
Deployment	**06/03/26**	**30/03/26**	**100%**	**21**
Server Setup	06/03	10/03	100%	4 days
Launch	30/03	30/03	100%	1 days



► DEVELOPMENT METHODOLOGY

- **Agile Scrum Framework**

The Agile Scrum framework was adopted to allow flexibility, continuous feedback, and effective handling of changing requirements within the limited project timeline

Sprint Structure

- Sprint Duration: 2 weeks
- Total Sprints: 5
- Sprint Planning: Monday (Week Start)
- Daily Standup: Every morning at 9:00 AM
- Sprint Review: Friday (Week End)
- Sprint Retrospective: Friday (After Review)

Agile Sprint planning Details

Sprint	Week	Focus Area	Deliverables
Sprint 1	3-4	Setup and Design	Environment setup, UI wireframes, Database design
Sprint 2	5-6	Courses and User profile	CRUD operations for Courses and User profile
Sprint 3	7-8	Assignment	File upload for students, Grading UI for lecturers
Sprint 4	9-10	Forums and Notifications	Discussion threads, Basic notification system.
Sprint 5	11-12	Integration and Polishing	Bug fixes, UAT, Final Deployment

Handling Challenges & Decisions (Case Study Scenarios)

Scenario 1: Scope Creep (Video Conferencing)

- **Situation:** Client requests video conferencing mid-project.

- **Response: Defer.**
    - *Reasoning:* Video streaming requires significant bandwidth, expensive infrastructure (WebRTC/Media Servers), and development time that does not fit the 3-month/₦5m constraint.
    - *Action:* Propose integrating a third-party link (Zoom/Google Meet API) as a "resource link" inside the course .

Scenario 2: Delay Risk (Developer Ill)

- **Situation:** One developer is down for 2 weeks in February.
- **Response: Resource Leveling.**
    - *Action:* The Project Manager (technical) steps in to assist with code reviews and minor bug fixes.
    - *Trade-off:* We reduce the scope of "Real-time notifications" from "In-app + SMS + Email" to just "Email" to save development hours. The sprint backlog is re-prioritized to ensure the "Assignment Submission" (Critical path) is not delayed.

Scenario 3: Budget Cut (20% Reduction)

- **Situation:** Budget drops from ₦5m to ₦4m.
- **Response: Cut "Should Have" Features.**
    - *Action:* Advanced discussion forum features are removed from the MVP, while a basic forum is retained to support essential student–lecturer interaction.. We switch from paid hosting tiers to free/tier-1 tiers for the pilot (e.g., using free credits on AWS/Azure for startups). We renegotiate team stipends or reduce hours slightly, focusing strictly on the *Core Learning Path* (Login -> View Content -> Submit Assignment -> Grade).

STAKEHOLDER COMMUNICATION & RECOMMENDATIONS

Budget Reduction Response: Stakeholder Communication

TO: Client Stakeholders, EduLearn Project Sponsors
FROM: Project Manager,EduLearn LMS
DATE: February 1, 2025
RE: Budget Adjustment & Revised Project Scope

Dear Stakeholders,

I am writing to address the recent 20% budget reduction (from ₦5 million to ₦4 million) and present our strategic approach to delivering maximum value within the revised financial constraints.

Budget Impact Analysis:
The reduction affects development hours (~400 hours), third-party service subscriptions, and QA scope. However, we remain committed to delivering a functional, valuable LMS that achieves your core educational objectives.

Proposed Phased Approach:
Phase 1 (Launch by March 31, 2025):

- ✓ User authentication for students, lecturers, and administrators
- ✓ Course management (creation, enrollment, content viewing)
- ✓ Assignment submission and grading system
- ✓ Essential discussion forums
- ✓ Basic email notifications

Phase 2 (Post-Launch, April-May 2025):

- ○ Real-time push notifications
- ○ Advanced forum features (file attachments, nested threading)
- ○ Analytics dashboard for administrators
- ○ Mobile responsiveness optimization

Best regards,

Project Management Team

APPENDIX

APPENDIX A: SYSTEM ARCHITECTURE OVERVIEW

The EduLearn LMS follows a three-tier web application architecture:
Presentation Layer (Frontend): Built using HTML, CSS, and JavaScript framework (e.g., React).
Application Layer (Backend): Implemented using Node.js (Express) or Django.
Data Layer (Database): MySQL or PostgreSQL used for persistent storage.

APPENDIX B: DATABASE DESIGN (SUMMARY)

Main entities include: Users, Courses, Enrollments, Assignments, Submissions, Grades, Forum Threads, Notifications.
Relationships: One Lecturer → Many Courses; One Course → Many Assignments; One Student → Many Submissions.

APPENDIX C: USER ROLES & PERMISSIONS

Students: Register, enroll in courses, submit assignments, join forums.
Lecturers: Create courses, upload materials, grade assignments, moderate forums.
Administrators: Manage users, manage courses, monitor system.

APPENDIX D: RISK REGISTER

Budget reduction – Scope adjustment and phased delivery.
Developer unavailability – Resource reallocation.
Server downtime – Cloud backups and monitoring.
Security breach – Role-based access control and encryption.

APPENDIX E: TECHNOLOGY STACK

Frontend: HTML, CSS, JavaScript (React)
Backend: Node.js (Express) or Django
Database: MySQL or PostgreSQL
Hosting: AWS or Azure
Version Control: Git & GitHub

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